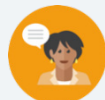


6.9 Proving Triangle Similarity – Part 2

Lesson Introduction

 Benchmark	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse Leg.		 Learning Target	I can use triangle similarity to prove that triangles are similar.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- What criteria are needed to prove two triangles are similar? - Do you need to know the measure of all sides or all angles to prove triangles similar?			
 Lesson Narrative	This lesson is an opportunity for students to apply the criteria for proving two triangles similar in writing a two-column proofs. These proofs are focused on overlapping and shared side triangles. Students will be challenged to use any reasoning learned in previous lessons and units to draw the conclusion.			
 Component Summary	6.9.1 Warm-Up (~5 min.)	Choosing between two future forms of transportation (<i>SE p. 338</i>)	6.9.2 Activity (35-40 min.)	Writing proofs of triangle similarity (<i>SE p. 338</i>)
	6.9.3 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 342</i>)		
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Similarity - Congruent - Proportional		(Optional) Chart Paper (Optional) Markers (Optional) Task Cards - Consider making multiple sets of the proofs regardless of implementing a Gallery Walk, stations, task cards, or other structure.	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may think there is only one way to order the statements in a proof.	Consider the different ways of grouping/pairing students for this lesson based on students' strengths and level of understanding. Depending on the choice of structure for 6.9.2 Activity, consider whether it is better to group the students homogeneously or heterogeneously. Consider also making a small teacher-led group if needed.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, Correct When reviewing the completed proofs from 6.9.2 Activity, provide students with an opportunity to analyze the completed work of other students. The intent is to prompt student reflection on the work of others to fortify their own reasoning and to engage students in meaningful conversation.	MA.K12.MTR.4.1: Discussion and Reflection 6.9.2 Activity is designed for students to discuss and reflect on a variety of proofs. Students will analyze the mathematical thinking of others and compare the efficiency of their process. Through this discussion, students develop skills in communicating mathematical ideas, methods, and vocabulary.
 Differentiate Instruction	6.9.2 Activity is an opportunity for students to access prior knowledge and practice the skill of writing proofs using different approaches. For students who struggle with proofs, consider assigning only proofs A-D or providing some of the placement of reasons and statements on proofs B, C, and F.	
Lesson Notes:		

6.9.1 Warm-Up: Would You Rather?

Component Overview: Students choose between two different options of futuristic vehicles and justify their choice.



6.9.1 Warm-Up: Would You Rather?

1. In 2021, General Motors introduced the eVTOL air-taxi and the PAV (personal autonomous vehicle), a self-driving car. Both are concepts with no known timeline for production.
 - a. Do you think a drone-like personal vehicle like the eVTOL air-taxi will be reality in your lifetime?
Answers will vary. Yes, people are making lots of cool stuff and this should be a great idea.
 - b. Which would you rather have as your vehicle, an air-taxi or a self-driving car, in the future?
Answers will vary. I would rather have a self-driving car.
 - c. Explain your choice.
Answers will vary. I think that self-driving vehicles are safer.



Consider doing research on the latest news on air-taxis, PAVs, and self-driving cars. Use what you learn to further engage students in conversations about the future of travel.

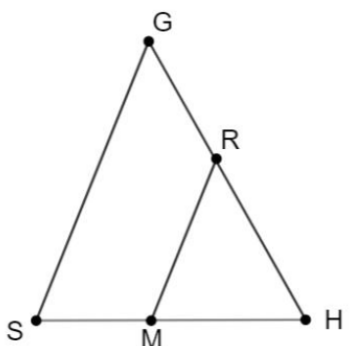
6.9.2 Activity: Proving Triangle Similarity

Component Overview: Students write formal two-column proofs justifying why two triangles are similar from given information.



6.9.2 Activity: Proving Triangle Similarity

With your partner, sort through the statement and reason cards. Then, write each proof in the tables provided.

Proof A	
Given: $\overline{GS} \parallel \overline{RM}$ Prove: $\triangle RHM \sim \triangle GHS$ 	
Statement	Reason
1. $\overline{GS} \parallel \overline{RM}$	1. Given
2. $\angle H \cong \angle H$	2. Reflexive Property of Congruence
3. $\angle GSH \cong \angle RMH$	3. If two parallel lines are intersected by a transversal, the corresponding angles are congruent
4. $\triangle RHM \sim \triangle GHS$	4. Angle-Angle Similarity Theorem



Consider using one of the following implementation options for this component:

- **Stations** – The proofs are stationed around the room with the accompanying statement and reason cards. Groups rotate through the stations, place the cards in the correct position, and then record their answers in their workbooks. Don't forget to remind the students to mix the cards up when leaving the station.
- **Gallery Walk** – The proofs are posted on poster paper around the room along with tape. ALL statement and reason cards are cut up, mixed up, and distributed to the student groups/pairs. For example, there are 48 cards, so a class of 24 students placed in pairs will be assigned four cards. Students rotate around the room and “build” the proofs by adding statements and reasons to the appropriate proof and on the appropriate lines.
- **Partner Pair Share** – Groups of two students are assigned three proofs, either Proofs A-C or Proofs D-F. After the pairs complete their proofs, they pair with another pair who completed the other set of proofs and explain to each other their assigned proofs.
- **Partner Switch** – Pairs of students work together to complete some or all of the proofs. Students then switch partners and verify each other's proofs.

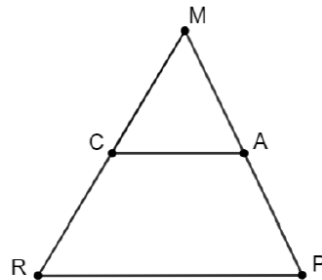
Any proofs not completed can be assigned for homework or as a station on another day.

6.9.2 Activity: Proving Triangle Similarity (continued)

Proof B

Given: $MC = \frac{2}{5}MR$ and
 $MA = \frac{2}{5}MP$

Prove: $\triangle MCA \sim \triangle MRP$



Statement	Reason
1. $MC = \frac{2}{5}MR$ and $MA = \frac{2}{5}MP$	1. Given
2. $\frac{MC}{MR} = \frac{2}{5}$, $\frac{MA}{MP} = \frac{2}{5}$	2. Division Property of Equality
3. $\frac{MC}{MR} = \frac{MA}{MP}$	3. Substitution Property
4. $\angle M \cong \angle M$	4. Reflexive Property of Congruence
5. $\triangle MCA \sim \triangle MRP$	5. Side-Angle-Side Similarity Theorem



Students may choose where to put statements showing the reflexive property anywhere in the proof prior to the conclusion statement.

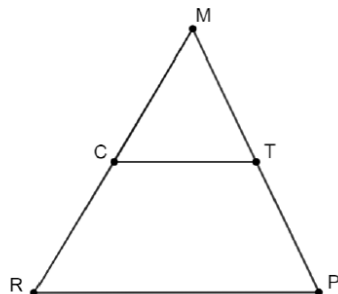
Students may interchange the substitution or transitive property reason statements in Proofs B, C, E, F, and G.

6.9.2 Activity: Proving Triangle Similarity (continued)

Proof C

Given: $\angle RPT$ and $\angle CTP$
are supplementary

Prove: $\triangle MCT \sim \triangle MRP$



As an extension problem, have students write this proof using a different set of statements and reasons. For example, this proof can also be written by proving $\overline{CT}$ is parallel to $\overline{RP}$ and then using angle pair relationships to establish congruence.

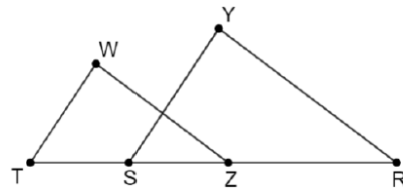
Statement	Reason
1. $\angle RPT$ and $\angle CTP$ are supplementary	1. Given
2. $m\angle RPT + m\angle CTP = 180^\circ$	2. Definition of Supplementary Angles
3. $\angle CTM$ and $\angle CTP$ are supplementary	3. If two angles form a linear pair, they are supplementary
4. $m\angle CTM + m\angle CTP = 180^\circ$	4. Definition of Supplementary Angles
5. $m\angle CTM + m\angle CTP = m\angle RPT + m\angle CTP$	5. Transitive Property of Equality
6. $m\angle CTM = m\angle RPT$	6. Subtraction Property of Equality
7. $\angle CTM \cong \angle RPT$	7. Definition of Congruence
8. $\angle M \cong \angle M$	8. Reflexive Property of Congruence
9. $\triangle MCT \sim \triangle MRP$	9. Angle-Angle Similarity Theorem

6.9.2 Activity: Proving Triangle Similarity (continued)

Proof D

Given: $\overline{WT} \parallel \overline{YS}$,
 $\overline{WZ} \parallel \overline{YR}$

Prove: $\triangle WZT \sim \triangle YRS$



Encourage students to think outside the box and draw the lines of each side well outside of the shape. This could help students see how $\overline{TR}$ is a transversal.

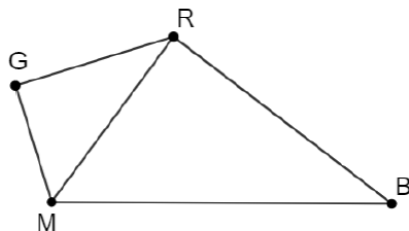
Statement	Reason
1. $\overline{WT} \parallel \overline{YS}$	1. Given
2. $\angle WTZ \cong \angle YSR$	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent
3. $\overline{WZ} \parallel \overline{YR}$	3. Given
4. $\angle WZT \cong \angle YRS$	4. If two parallel lines are intersected by a transversal, then corresponding angles are congruent
5. $\triangle WZT \sim \triangle YRS$	5. Angle-Angle Similarity Theorem

6.9.2 Activity: Proving Triangle Similarity (continued)

Proof E

Given: $\therefore GM = \frac{3}{5}RM$,
 $RG = \frac{3}{5}BR$, $RM = \frac{3}{5}BM$

Prove:
 $\triangle MGR \sim \triangle MRB$



Although this is a short proof, students may need more support on this proof because the statements are algebraically based.

Statement	Reason
1. $GM = \frac{3}{5}RM$, $RG = \frac{3}{5}BR$, $RM = \frac{3}{5}BM$	1. Given
2. $\frac{GM}{RM} = \frac{3}{5}$, $\frac{RG}{BR} = \frac{3}{5}$, $\frac{RM}{BM} = \frac{3}{5}$	2. Division Property of Equality
3. $\frac{GM}{RM} = \frac{RG}{BR} = \frac{RM}{BM}$	3. Substitution Property
4. $\triangle MGR \sim \triangle MRB$	4. Side-Side-Side Similarity Theorem

6.9.2 Activity: Proving Triangle Similarity (continued)

Proof F	
Given: $\overline{WT} \perp \overline{VT}$ and $\overline{WV} \perp \overline{RT}$ Prove: $\triangle RTW \sim \triangle TVW$	
Statement	Reason
1. $\overline{WT} \perp \overline{VT}$	1. Given
2. $\angle VTW$ is a right angle	2. Definition of Perpendicular Lines
3. $m\angle VTW = 90^\circ$	3. Definition of Right Angle
4. $\overline{WV} \perp \overline{RT}$	4. Given
5. $\angle TRW$ is a right angle	5. Definition of Perpendicular Lines
6. $m\angle TRW = 90^\circ$	6. Definition of Right Angle
7. $m\angle TRW = m\angle VTW$	7. Substitution Property
8. $\angle TRW \cong \angle VTW$	8. Definition of Congruence
9. $\angle W \cong \angle W$	9. Reflexive Property of Congruence
10. $\triangle RTW \sim \triangle TVW$	10. Angle-Angle Similarity Theorem



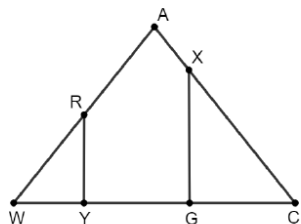
The students may struggle with the overlapping of the triangles in this image. This proof may be more manageable if students separate the triangles by copying each onto another piece of paper.

6.9.2 Activity: Proving Triangle Similarity (continued)

Proof G

Given: $\overline{WA} \cong \overline{CA}$,
 $\overline{RY} \perp \overline{WC}$, $\overline{XG} \perp \overline{WC}$

Prove: $\triangle RYW \sim \triangle XGC$



Statement	Reason
1. $\overline{WA} \cong \overline{CA}$	1. Given
2. $WA = CA$	2. Definition of Congruence
3. $\triangle WAC$ is Isosceles	3. Definition of Isosceles Triangles
4. $m\angle AWC = m\angle ACW$	4. Definition of Isosceles Triangles
5. $\angle AWC \cong \angle ACW$	5. Definition of Congruence
6. $\overline{RY} \perp \overline{WC}$	6. Given
7. $\angle RYW$ is a right angle	7. Definition of Perpendicular Lines
8. $m\angle RYW = 90^\circ$	8. Definition of Right Angle
9. $\overline{XG} \perp \overline{WC}$	9. Given
10. $\angle XGC$ is a right angle	10. Definition of Perpendicular Lines
11. $m\angle XGC = 90^\circ$	11. Definition of Right Angle
12. $m\angle RYW = m\angle XGC$	12. Substitution Property
13. $\angle RYW \cong \angle XGC$	13. Definition of Congruence
14. $\triangle RYW \sim \triangle XGC$	14. Angle-Angle Similarity Theorem



While students may choose to write all the given statements at the beginning of the proof, encourage them to write only one given statement and then the statements that can be concluded from that one. This move will help students make connections if they still see proofs as disconnected phrases.



For students who finish early, ask them to name the transformations that would show the two triangles are similar. For Honors Geometry, this would be a good segue into Lesson H3 in Unit 6.

6.9.3 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



6.9.3 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about proving triangle similarity.

Answers will vary.